

Lesson 8 — Sensor Triggers (Distance & Color)

Matches LEGO Teacher Portal lesson: "Spin a Song".

Objective (I can...): I can make my model react when a sensor detects something. **CS concept:** Sensor-driven events · **Builds on:** Lesson 7 (button/tap inputs).

AI Connection (2 min): A self-driving car's camera "seeing" a stop sign is a sensor event, just like your color sensor spotting a card. *Impacts prompt:* "Who does a self-driving car help? Could it bother or worry anyone?"

Materials: LEGO CS & AI kit + device per pair, a small object or color card to trigger the sensor. **Model reuse:** Events model from Lesson 6; add/attach the sensor as a small reconfig if needed — no full rebuild.

60-minute flow - Warm-up (8): "Sensors around us" — pairs name real sensors (automatic doors, hand dryers) and what each detects. - **Concept + demo (10):** Show a distance or color sensor block firing an event when an object nears or a color appears. - **Build/Code (30):** Pairs program the model so an action happens when the sensor detects an object/color (hand near → light on). *Swap at 15 min.* - **Challenge (7):** Change the trigger distance or color; test how sensitivity changes behavior. - **Share + pack-up (5):** Pairs demonstrate the sensor triggering their action. Model stays built.

Pair roles: Driver codes, Navigator operates the trigger object and observes. Swap at 15 min. (*Trio rotates code/trigger/observe.*)

Differentiation: *Support* — sentence-framed pseudocode ("WHEN the sensor sees , **THEN** ") to fill in. *Extension / fast-finisher* — two sensor triggers doing different things. *Reteach* — watch the sensor's live reading on screen while moving the object closer.

Exit check: Pair's model reliably reacts to the sensor event, not to Run. **At-home (optional):** Spot 3 sensors at home; write what each detects. ~10 min.
